ROCSCIENCE

Case12_Water_Surcharge
SLIDE - An Interactive Slope Stability Program
Rocscience Inc.
Date Created: 12/12/2016
Software Version: 9.041

Table of Contents

Project Summary 3
General Settings 4
Analysis Options 5
Groundwater Analysis 6
Random Numbers 7
Surface Options 8
Seismic Loading 9
Loading 10
Materials 11
Global Minimums 12
Method: bishop simplified 12
Method: spencer 12
Valid and Invalid Surfaces 13
Method: bishop simplified 13
Method: spencer 13
Error Code Descriptions 13
Slice Data 14
Global Minimum Query (bishop simplified) - Safety Factor: 0.311775 14
Global Minimum Query (spencer) - Safety Factor: 0.316117 16
Interslice Data 18
Global Minimum Query (bishop simplified) - Safety Factor: 0.311775 18
Global Minimum Query (spencer) - Safety Factor: 0.316117 19
Entity Information 20
Group 1 20
Shared Entities 20
Scenario-based Entities 20

Slide2 Analysis Information

Case12_Water_Surcharge

Project Summary

File Name:	Case12_Water_Surcharge.slmd
Slide2 Modeler Version:	9.041
Compute Time:	00h:00m:01.743s
Project Title:	SLIDE - An Interactive Slope Stability Program
Author:	Rocscience Inc.
Company:	Rocscience Inc.
Date Created:	12/12/2016

General Settings

Units of Measurement:	Metric Units
Time Units:	days
Permeability Units:	meters/second
Data Output:	Standard
Failure Direction:	Right to Left

Analysis Options

Slices Type:	Vertical
Analysis Methods Used	
	Bishop simplified
	Spencer
Number of slices:	50
Tolerance:	0.005
Maximum number of iterations:	75
Check malpha < 0.2:	Yes
Create Interslice boundaries at intersections with water tables and piezos:	Yes
Initial trial value of FS:	1
Steffensen Iteration:	Yes

Groundwater Analysis

Groundwater Method:	Water Surfaces
Pore Fluid Unit Weight [kN/m3]:	9.81
Advanced Groundwater Method:	None

Random Numbers

Pseudo-random Seed:	10116
Random Number Generation Method:	Park and Miller v.3

Surface Options

Surface Type:	Circular
Search Method:	Auto Refine Search
Divisions along slope:	20
Circles per division:	10
Number of iterations:	10
Divisions to use in next iteration:	50%
Composite Surfaces:	Disabled
Minimum Elevation:	Not Defined
Minimum Depth:	Not Defined
Minimum Area:	Not Defined
Minimum Weight:	Not Defined

Seismic Loading

Advanced seismic analysis:	No
Staged pseudostatic analysis:	No

Loading

1 Distributed Load present

Distributed Load 1	
Distribution:	Constant
Magnitude [kPa]:	100
Orientation:	Normal to boundary

Materials

top layer	
Color	
Strength Type	Mohr-Coulomb
Unit Weight	18.82 kN/m3
Cohesion	29.4 kPa
Phi	12 °
Water Surface	Water Table
Hu Type	Automatically Calculated
Specify alternate strength type above water surface	No

middle layer	
Color	
Strength Type	Mohr-Coulomb
Unit Weight	18.82 kN/m3
Cohesion	9.8 kPa
Phi	5 °
Water Surface	Water Table
Hu Type	Automatically Calculated
Specify alternate strength type above water surface	No

lower layer	
Color	
Strength Type	Mohr-Coulomb
Unit Weight	18.82 kN/m3
Cohesion	294 kPa
Phi	40 °
Water Surface	Water Table
Hu Type	Automatically Calculated
Specify alternate strength type above water surface	No

Global Minimums

Method: bishop simplified

FS	0.311775
Center:	27.106, 45.138
Radius:	31.409
Left Slip Surface Endpoint:	18.080, 15.053
Right Slip Surface Endpoint:	56.834, 35.000
Left Slope Intercept:	18.080 19.000
Right Slope Intercept:	56.834 35.000
Resisting Moment:	30067.7 kN-m
Driving Moment:	96440.6 kN-m
Total Slice Area:	363.218 m2
Surface Horizontal Width:	38.7549 m
Surface Average Height:	9.37219 m

Method: spencer

FS	0.316117
Center:	27.106, 45.138
Radius:	31.409
Left Slip Surface Endpoint:	18.080, 15.053
Right Slip Surface Endpoint:	56.834, 35.000
Left Slope Intercept:	18.080 19.000
Right Slope Intercept:	56.834 35.000
Resisting Moment:	30486.5 kN-m
Driving Moment:	96440.6 kN-m
Resisting Horizontal Force:	766.759 kN
Driving Horizontal Force:	2425.55 kN
Total Slice Area:	363.218 m2
Surface Horizontal Width:	38.7549 m
Surface Average Height:	9.37219 m

Global Minimum Support Data

No Supports Present

Valid and Invalid Surfaces

Method: bishop simplified

Number of Valid Surfaces:	9951
Number of Invalid Surfaces:	0

Method: spencer

Number of Valid Surfaces:	7694
Number of Invalid Surfaces:	2257

Error Codes

Error Code -108 reported for 290 surfaces
Error Code -111 reported for 1964 surfaces
Error Code -112 reported for 3 surfaces

Error Code Descriptions

The following errors were encountered during the computation:
-108 = Total driving moment or total driving force < 0.1. This is to limit the calculation of extremely high safety factors if the driving force is very small (0.1 is an arbitrary number).
-111 = Safety factor equation did not converge
-112 = The coefficient M-Alpha = cos(alpha)(1+tan(alpha)tan(phi)/F) < 0.2 for the final iteration of the safety factor calculation. This screens out some slip surfaces which may not be valid in the context of the analysis, in particular, deep seated slip surfaces with many high negative base angle slices in the passive zone.

Slice Data

Global Minimum Query (bishop simplified) - Safety Factor: 0.311775

Slice Number	Width [m]	Weight [kN]	Angle of Slice Base [deg]	Base Material	Base Cohesion [kPa]	Base Friction Angle [deg]	Shear Stress [kPa]	Shear Strength [kPa]	Base Normal Stress [kPa]	Pore Pressure [kPa]	Effective Normal Stress [kPa]	Base Vertical Stress [kPa]	Effective Vertical Stress [kPa]
1	0.76578	32.9918	-15.9745	middle layer	9.8	5	35.1384	10.9553	52.9996	39.795	13.2046	42.9408	3.1458
2	0.76578	39.5237	-14.5264	middle layer	9.8	5	36.8132	11.4774	61.0165	41.8435	19.173	51.4779	9.63435
3	0.76578	45.7588	-13.0878	middle layer	9.8	5	38.4115	11.9757	68.5587	43.69	24.8687	59.6286	15.9386
4	0.76578	51.7027	-11.6575	middle layer	9.8	5	39.936	12.451	75.6396	45.3382	30.3014	67.4001	22.0619
5	0.76578	57.3599	-10.2346	middle layer	9.8	5	41.3892	12.9041	82.2714	46.7913	35.4801	74.7985	28.0072
6	0.76578	62.7347	-8.818	middle layer	9.8	5	42.7732	13.3356	88.4645	48.0522	40.4123	81.8291	33.7769
7	0.76578	67.8305	-7.40683	middle layer	9.8	5	44.09	13.7462	94.2278	49.1232	45.1046	88.4962	39.373
8	0.76578	72.7888	-6.00017	middle layer	9.8	5	45.3933	14.1525	99.7557	50.0063	49.7494	94.9846	44.9783
9	0.76578	80.1446	-4.59713	middle layer	9.8	5	49.0829	15.3028	108.548	45.6509	62.8976	104.602	58.951
10	0.76578	88.2541	-3.19685	middle layer	9.8	5	50.8962	15.8681	118.05	48.6908	69.3592	115.207	66.5165
11	0.76578	96.0932	-1.79848	middle layer	9.8	5	52.6309	16.409	127.113	51.5722	75.5412	125.461	73.8886
12	0.76578	103.663	-0.401176	middle layer	9.8	5	54.2884	16.9258	135.743	54.2956	81.4476	135.363	81.0674
13	0.76578	110.963	0.995885	middle layer	9.8	5	55.8697	17.4188	143.944	56.8613	87.0825	144.915	88.0537
14	0.76578	117.994	2.39354	middle layer	9.8	5	57.3757	17.8883	151.719	59.2692	92.4499	154.117	94.8482
15	0.76578	124.755	3.79262	middle layer	9.8	5	58.807	18.3345	159.069	61.5189	97.5497	162.967	101.448
16	0.76578	131.245	5.19397	middle layer	9.8	5	60.164	18.7576	165.997	63.6098	102.387	171.465	107.856
17	0.76578	137.463	6.59845	middle layer	9.8	5	61.447	19.1577	172.5	65.541	106.959	179.608	114.067
18	0.76578	143.406	8.00692	middle layer	9.8	5	62.6562	19.5346	178.578	67.3113	111.267	187.392	120.08
19	0.76578	149.072	9.42028	middle layer	9.8	5	63.7915	19.8886	184.232	68.9191	115.313	194.816	125.897
20	0.76578	154.458	10.8395	middle layer	9.8	5	64.8525	20.2194	189.457	70.3626	119.094	201.875	131.512
21	0.76578	159.559	12.2654	middle layer	9.8	5	65.8391	20.527	194.249	71.6397	122.61	208.563	136.923
22	0.76578	164.372	13.6991	middle layer	9.8	5	66.7504	20.8111	198.605	72.7476	125.857	214.876	142.128
23	0.76578	168.891	15.1416	middle layer	9.8	5	67.5853	21.0714	202.517	73.6835	128.833	220.805	147.122
24	0.76578	173.111	16.594	middle layer	9.8	5	68.3435	21.3078	205.978	74.444	131.534	226.345	151.901
25	1.15665	268.739	18.4349	middle layer	9.8	5	69.1742	21.5668	209.61	75.1147	134.495	232.668	157.553
26	0.783382	186.568	20.3089	middle layer	9.8	5	69.9043	21.7944	212.651	75.5547	137.096	238.521	162.967
27	0.783382	189.816	21.8406	middle layer	9.8	5	70.3895	21.9457	214.489	75.6635	138.825	242.701	167.037
28	0.783382	192.704	23.3889	middle layer	9.8	5	70.7863	22.0694	215.806	75.5658	140.24	246.421	170.855
29	0.783382	195.218	24.9555	middle layer	9.8	5	71.0917	22.1646	216.583	75.2545	141.328	249.666	174.411
30	0.783382	197.346	26.5423	middle layer	9.8	5	71.3027	22.2304	216.802	74.7215	142.081	252.418	177.697

31	0.783382	199.071	28.1514	middle layer	9.8	5	71.4159	22.2657	216.441	73.9578	142.483	254.656	180.698
32	0.783382	200.376	29.7851	middle layer	9.8	5	71.4272	22.2692	215.476	72.9531	142.523	256.358	183.405
33	0.783382	201.239	31.4459	middle layer	9.8	5	71.3319	22.2395	213.879	71.6957	142.184	257.499	185.803
34	0.783382	201.637	33.1368	middle layer	9.8	5	71.1244	22.1748	211.617	70.1721	141.445	258.048	187.876
35	0.783382	201.545	34.8609	middle layer	9.8	5	70.7988	22.0733	208.651	68.367	140.284	257.969	189.602
36	0.783382	200.929	36.622	middle layer	9.8	5	70.3478	21.9327	204.939	66.2627	138.677	257.226	190.963
37	0.783382	199.756	38.4243	middle layer	9.8	5	69.7625	21.7502	200.43	63.8385	136.592	255.771	191.933
38	0.783382	197.982	40.2729	middle layer	9.8	5	69.0331	21.5228	195.063	61.0704	133.993	253.552	192.481
39	0.783382	193.431	42.1736	middle layer	9.8	5	84.2159	26.2564	246.027	57.9296	188.098	322.319	264.389
40	0.783382	182.85	44.1333	middle layer	9.8	5	86.5117	26.9722	250.662	54.3821	196.279	334.595	280.213
41	0.783382	171.234	46.1605	middle layer	9.8	5	82.8296	25.8242	233.544	50.3865	183.158	319.799	269.412
42	0.783382	158.747	48.2655	middle layer	9.8	5	78.9848	24.6255	215.347	45.892	169.455	303.891	257.999
43	0.783382	145.278	50.4614	middle layer	9.8	5	74.9541	23.3688	195.927	40.8348	155.092	286.729	245.894
44	0.783382	130.683	52.7647	middle layer	9.8	5	70.7084	22.0451	175.095	35.1334	139.962	268.131	232.998
45	0.783382	114.777	55.1976	middle layer	9.8	5	66.208	20.642	152.605	28.6802	123.924	247.857	219.177
46	0.783382	97.3022	57.7902	middle layer	9.8	5	61.3995	19.1428	128.118	21.3284	106.789	225.582	204.253
47	0.702508	70.852	60.4293	top layer	29.4	12	101.674	31.6995	24.1887	13.3703	10.8184	203.381	190.011
48	0.702508	53.4951	63.1464	top layer	29.4	12	71.1066	22.1693	-29.3509	4.66699	-34.0179	111.089	106.422
49	0.681737	33.1432	66.0999	top layer	29.4	12	50.6372	15.7874	-64.0426	0	-64.0426	50.2263	50.2263
50	0.681737	11.637	69.4026	top layer	29.4	12	37.9905	11.8445	-82.592	0	-82.592	18.4943	18.4943

Global Minimum Query (spencer) - Safety Factor: 0.316117

Slice Number	Width [m]	Weight [kN]	Angle of Slice Base [deg]	Base Material	Base Cohesion [kPa]	Base Friction Angle [deg]	Shear Stress [kPa]	Shear Strength [kPa]	Base Normal Stress [kPa]	Pore Pressure [kPa]	Effective Normal Stress [kPa]	Base Vertical Stress [kPa]	Effective Vertical Stress [kPa]
1	0.76578	32.9918	-15.9745	middle layer	9.8	5	40.8906	12.9262	75.5276	39.795	35.7326	63.8221	24.0271
2	0.76578	39.5237	-14.5264	middle layer	9.8	5	42.3596	13.3906	82.8841	41.8435	41.0406	71.9083	30.0648
3	0.76578	45.7588	-13.0878	middle layer	9.8	5	43.7121	13.8181	89.6175	43.69	45.9275	79.4552	35.7652
4	0.76578	51.7027	-11.6575	middle layer	9.8	5	44.9563	14.2115	95.7614	45.3382	50.4232	86.4861	41.1479
5	0.76578	57.3599	-10.2346	middle layer	9.8	5	46.0996	14.5729	101.345	46.7913	54.554	93.022	46.2307
6	0.76578	62.7347	-8.818	middle layer	9.8	5	47.1483	14.9044	106.395	48.0522	58.3433	99.0814	51.0292
7	0.76578	67.8305	-7.40683	middle layer	9.8	5	48.108	15.2078	110.934	49.1232	61.8111	104.68	55.5571
8	0.76578	72.7888	-6.00017	middle layer	9.8	5	49.05	15.5055	115.221	50.0063	65.2147	110.065	60.0591
9	0.76578	80.1446	-4.59713	middle layer	9.8	5	52.7065	16.6614	124.077	45.6509	78.4262	119.839	74.1882
10	0.76578	88.2541	-3.19685	middle layer	9.8	5	54.4124	17.2007	133.281	48.6908	84.5901	130.242	81.551
11	0.76578	96.0932	-1.79848	middle layer	9.8	5	56.0031	17.7035	141.909	51.5722	90.3372	140.151	88.5787
12	0.76578	103.663	-0.401176	middle layer	9.8	5	57.4834	18.1715	149.982	54.2956	95.6863	149.579	95.2838
13	0.76578	110.963	0.995885	middle layer	9.8	5	58.8574	18.6058	157.512	56.8613	100.651	158.535	101.674
14	0.76578	117.994	2.39354	middle layer	9.8	5	60.129	19.0078	164.515	59.2692	105.245	167.028	107.759
15	0.76578	124.755	3.79262	middle layer	9.8	5	61.3015	19.3784	171.001	61.5189	109.483	175.065	113.546
16	0.76578	131.245	5.19397	middle layer	9.8	5	62.3777	19.7187	176.981	63.6098	113.371	182.651	119.041
17	0.76578	137.463	6.59845	middle layer	9.8	5	63.3604	20.0293	182.463	65.541	116.922	189.792	124.251
18	0.76578	143.406	8.00692	middle layer	9.8	5	64.2518	20.3111	187.453	67.3113	120.142	196.491	129.18
19	0.76578	149.072	9.42028	middle layer	9.8	5	65.0538	20.5646	191.959	68.9191	123.04	202.752	133.833
20	0.76578	154.458	10.8395	middle layer	9.8	5	65.7681	20.7904	195.984	70.3626	125.621	208.577	138.214
21	0.76578	159.559	12.2654	middle layer	9.8	5	66.396	20.9889	199.53	71.6397	127.89	213.964	142.325
22	0.76578	164.372	13.6991	middle layer	9.8	5	66.9391	21.1606	202.6	72.7476	129.853	218.917	146.169
23	0.76578	168.891	15.1416	middle layer	9.8	5	67.3981	21.3057	205.194	73.6835	131.511	223.432	149.749
24	0.76578	173.111	16.594	middle layer	9.8	5	67.7736	21.4244	207.311	74.444	132.867	227.508	153.064
25	1.15665	268.739	18.4349	middle layer	9.8	5	68.1181	21.5333	209.227	75.1147	134.112	231.933	156.818
26	0.783382	186.568	20.3089	middle layer	9.8	5	68.3525	21.6074	210.515	75.5547	134.96	235.811	160.257
27	0.783382	189.816	21.8406	middle layer	9.8	5	68.4376	21.6343	210.931	75.6635	135.267	238.36	162.697
28	0.783382	192.704	23.3889	middle layer	9.8	5	68.4361	21.6338	210.826	75.5658	135.26	240.425	164.859
29	0.783382	195.218	24.9555	middle layer	9.8	5	68.3468	21.6056	210.193	75.2545	134.938	241.999	166.744
30	0.783382	197.346	26.5423	middle layer	9.8	5	68.1691	21.5494	209.018	74.7215	134.297	243.069	168.348
31	0.783382	199.071	28.1514	middle layer	9.8	5	67.9018	21.4649	207.288	73.9578	133.331	243.623	169.665
32	0.783382	200.376	29.7851	middle layer	9.8	5	67.5437	21.3517	204.989	72.9531	132.036	243.648	170.695
33	0.783382	201.239	31.4459	middle layer	9.8	5	67.0922	21.209	202.102	71.6957	130.406	243.129	171.433

34	0.783382	201.637	33.1368	middle layer	9.8	5	66.5456	21.0362	198.602	70.1721	128.43	242.044	171.871
35	0.783382	201.545	34.8609	middle layer	9.8	5	65.9009	20.8324	194.468	68.367	126.101	240.374	172.007
36	0.783382	200.929	36.622	middle layer	9.8	5	65.1547	20.5965	189.668	66.2627	123.405	238.095	171.832
37	0.783382	199.756	38.4243	middle layer	9.8	5	64.3034	20.3274	184.167	63.8385	120.329	235.178	171.339
38	0.783382	197.982	40.2729	middle layer	9.8	5	63.3421	20.0235	177.925	61.0704	116.854	231.591	170.521
39	0.783382	193.431	42.1736	middle layer	9.8	5	76.4005	24.1515	221.968	57.9296	164.038	291.179	233.25
40	0.783382	182.85	44.1333	middle layer	9.8	5	77.9971	24.6562	224.189	54.3821	169.807	299.862	245.479
41	0.783382	171.234	46.1605	middle layer	9.8	5	74.4022	23.5198	207.205	50.3865	156.818	284.683	234.297
42	0.783382	158.747	48.2655	middle layer	9.8	5	70.7412	22.3625	189.483	45.892	143.591	268.785	222.893
43	0.783382	145.278	50.4614	middle layer	9.8	5	67.0068	21.182	170.932	40.8348	130.097	252.106	211.271
44	0.783382	130.683	52.7647	middle layer	9.8	5	63.1889	19.9751	151.435	35.1334	116.302	234.577	199.444
45	0.783382	114.777	55.1976	middle layer	9.8	5	59.2747	18.7377	130.839	28.6802	102.158	216.116	187.436
46	0.783382	97.3022	57.7902	middle layer	9.8	5	55.247	17.4645	108.934	21.3284	87.606	196.632	175.303
47	0.702508	70.852	60.4293	top layer	29.4	12	103.972	32.8672	29.6822	13.3703	16.3119	212.923	199.553
48	0.702508	53.4951	63.1464	top layer	29.4	12	77.8223	24.6009	-17.9108	4.66699	-22.5778	135.793	131.126
49	0.681737	33.1432	66.0999	top layer	29.4	12	60.5703	19.1473	-48.2351	0	-48.2351	88.4491	88.4491
50	0.681737	11.637	69.4026	top layer	29.4	12	50.5775	15.9884	-63.0963	0	-63.0963	71.4817	71.4817

Interslice Data

Global Minimum Query (bishop simplified) - Safety Factor: 0.311775

Slice Number	X coordinate [m]	Y coordinate - Bottom [m]	Interslice Normal Force [kN]	Interslice Shear Force [kN]	Interslice Force Angle [deg]
1	18.0795	15.053	76.4132	0	0
2	18.8453	14.8338	56.6363	0	0
3	19.6111	14.6354	112.469	0	0
4	20.3769	14.4574	167.05	0	0
5	21.1427	14.2994	219.97	0	0
6	21.9084	14.1611	270.855	0	0
7	22.6742	14.0423	319.363	0	0
8	23.44	13.9428	365.179	0	0
9	24.2058	13.8623	408.163	0	0
10	24.9716	13.8007	451.904	0	0
11	25.7373	13.7579	495.379	0	0
12	26.5031	13.7339	538.172	0	0
13	27.2689	13.7285	579.887	0	0
14	28.0347	13.7418	620.151	0	0
15	28.8005	13.7738	658.613	0	0
16	29.5662	13.8246	694.937	0	0
17	30.332	13.8942	728.805	0	0
18	31.0978	13.9828	759.916	0	0
19	31.8636	14.0905	787.985	0	0
20	32.6294	14.2176	812.739	0	0
21	33.3951	14.3642	833.923	0	0
22	34.1609	14.5307	851.291	0	0
23	34.9267	14.7173	864.615	0	0
24	35.6925	14.9246	873.675	0	0
25	36.4583	15.1528	878.269	0	0
26	37.6149	15.5383	876.337	0	0
27	38.3983	15.8282	868.675	0	0
28	39.1817	16.1422	855.696	0	0
29	39.9651	16.481	837.248	0	0
30	40.7484	16.8456	813.199	0	0
31	41.5318	17.2369	783.434	0	0
32	42.3152	17.6561	747.861	0	0
33	43.0986	18.1044	706.413	0	0
34	43.882	18.5835	659.048	0	0
35	44.6653	19.0949	605.76	0	0
36	45.4487	19.6406	546.58	0	0
37	46.2321	20.2228	481.585	0	0
38	47.0155	20.8443	410.91	0	0
39	47.7989	21.508	334.76	0	0
40	48.5823	22.2177	225.205	0	0
41	49.3656	22.9777	101.511	0	0
42	50.149	23.7935	-25.0354	0	0
43	50.9324	24.6717	-153.147	0	0
44	51.7158	25.6207	-281.194	0	0
45	52.4992	26.6514	-407.062	0	0
46	53.2826	27.7785	-527.918	0	0
47	54.0659	29.022	-639.813	0	0
48	54.7684	30.2601	-599.341	0	0
49	55.471	31.6476	-509.368	0	0
50	56.1527	33.186	-376.809	0	0
51	56.8344	35	0	0	0

Global Minimum Query (spencer) - Safety Factor: 0.316117

Slice Number	X coordinate [m]	Y coordinate - Bottom [m]	Interslice Normal Force [kN]	Interslice Shear Force [kN]	Interslice Force Angle [deg]
1	18.0795	15.053	76.4132	0	0
2	18.8453	14.8338	66.3172	16.1237	13.6652
3	19.6111	14.6354	131.09	31.872	13.6652
4	20.3769	14.4574	193.849	47.1307	13.6652
5	21.1427	14.2994	254.178	61.7985	13.6652
6	21.9084	14.1611	311.707	75.7856	13.6652
7	22.6742	14.0423	366.109	89.0122	13.6652
8	23.44	13.9428	417.092	101.408	13.6652
9	24.2058	13.8623	464.561	112.949	13.6652
10	24.9716	13.8007	512.509	124.606	13.6651
11	25.7373	13.7579	559.822	136.11	13.6652
12	26.5031	13.7339	606.063	147.352	13.6652
13	27.2689	13.7285	650.828	158.236	13.6652
14	28.0347	13.7418	693.743	168.67	13.6652
15	28.8005	13.7738	734.461	178.57	13.6652
16	29.5662	13.8246	772.662	187.858	13.6652
17	30.332	13.8942	808.046	196.461	13.6652
18	31.0978	13.9828	840.338	204.312	13.6652
19	31.8636	14.0905	869.283	211.349	13.6652
20	32.6294	14.2176	894.645	217.515	13.6652
21	33.3951	14.3642	916.205	222.757	13.6652
22	34.1609	14.5307	933.764	227.027	13.6652
23	34.9267	14.7173	947.138	230.278	13.6652
24	35.6925	14.9246	956.161	232.472	13.6652
25	36.4583	15.1528	960.683	233.571	13.6652
26	37.6149	15.5383	958.699	233.089	13.6652
27	38.3983	15.8282	951.142	231.252	13.6652
28	39.1817	16.1422	938.456	228.167	13.6652
29	39.9651	16.481	920.564	223.817	13.6652
30	40.7484	16.8456	897.408	218.187	13.6652
31	41.5318	17.2369	868.95	211.268	13.6652
32	42.3152	17.6561	835.178	203.057	13.6652
33	43.0986	18.1044	796.108	193.558	13.6652
34	43.882	18.5835	751.782	182.781	13.6652
35	44.6653	19.0949	702.279	170.745	13.6652
36	45.4487	19.6406	647.714	157.479	13.6652
37	46.2321	20.2228	588.252	143.022	13.6652
38	47.0155	20.8443	524.11	127.427	13.6652
39	47.7989	21.508	455.572	110.764	13.6653
40	48.5823	22.2177	357.819	86.9968	13.6652
41	49.3656	22.9777	248.449	60.4054	13.6652
42	50.149	23.7935	137.624	33.4606	13.6652
43	50.9324	24.6717	26.5674	6.45934	13.6652
44	51.7158	25.6207	-83.227	-20.235	13.6652
45	52.4992	26.6514	-189.883	-46.1664	13.6652
46	53.2826	27.7785	-290.971	-70.7439	13.6652
47	54.0659	29.022	-383.211	-93.1702	13.6652
48	54.7684	30.2601	-347.017	-84.3704	13.6652
49	55.471	31.6476	-267.568	-65.0538	13.6652
50	56.1527	33.186	-152.124	-36.9859	13.6652
51	56.8344	35	0	0	0

Discharge Sections

Entity Information

Group 1

Shared Entities

Type	Coordinates (x,y)
External Boundary	0, 3 96, 3 96, 35 72, 35 48, 35 24, 19 18, 15 0, 15
Material Boundary	24, 19 72, 35
Material Boundary	0, 3 96, 35

Scenario-based Entities

Type	Coordinates (x,y)	Master Scenario
Water Table	0, 19 24, 19 56.976, 32.2524 96, 32.2524	Assigned to: top layer middle layer lower layer
Distributed Load	55, 35 48, 35	Constant DistributionOrientation: Normal to boundaryMagnitude: 100 kN/m2Creates Excess Pore Pressure: No